Analog Function Generator

Designed by **uSquad**

PRODUCT DESCRIPTION

The **uSquad Analog Function Generator** is a precision laboratory instrument engineered for educational and experimental applications.

Utilizing a purely analog signal processing chain, it delivers high-fidelity waveforms without the quantization noise associated with digital synthesis. Key capabilities include a wide frequency range of 20Hz to 20kHz and a robust output amplitude of up to 24Vpp (-12V to +12V).

This device is ideal for student laboratories, analyzing analog circuits, and demonstrating fundamental signal generation principles such as Wien Bridge oscillation and Schmitt Trigger hysteresis.

Figure 1: Product Overview

KEY FEATURES

- **Versatile Waveform Output:** Generates Sine, Square, Triangular, Sawtooth, and PWM signals.
- **High-Fidelity Audio Range:** Covers 20 Hz – 20 kHz in four user-selectable bands.
- **High-Power Output Stage:** Capable of driving loads $\geq 50\Omega$ with ± 12V rail-to-rail swing.
- **Variable Duty Cycle:** Full 1% to 99% PWM control for motor driving and signaling commands.
- **Mains Powered:** Integrated 230V AC linear power supply ensures low-noise operation.
- **Durable Construction:** Custom PLA enclosure with industrial-grade FR4 PCB.

ELECTRICAL SPECIFICATIONS

Unless otherwise noted, $T_A = 25°C$, Main Supply = 230V AC.

Parameter	Min	Typ	Max	Unit
Power Input				
Input Voltage (AC)	200	230	240	V
Internal Rails	-	± 12	-	V
Signal Output				
Output Amplitude	-12	-	+12	V
Frequency Range	20	-	20,000	Hz
Output Impedance	-	50	-	Ω
PWM Duty Cycle	1	-	99	%

CONFIGURATION

The device utilizes precise capacitor banks for frequency band selection.

Square / Triangle Mode

Range	Capacitor
20 Hz – 130 Hz	10 μF
130 Hz – 1.5 kHz	1 μF
1.5 kHz – 15.6 kHz	68 nF
15.6 kHz – 20 kHz	10 nF

Sine Wave Mode (Wien Bridge)

Range	Capacitor
20 Hz – 690 Hz	100 nF
690 Hz – 5.5 kHz	10 nF
5.5 kHz – 12 kHz	1 nF
12 kHz – 20 kHz	330 pF

MECHANICAL SPECIFICATIONS

Build Materials

- **PCB Stackup:** 2-Layer FR4, 1oz Copper.
- **Enclosure:** 3D Printed Polylactic Acid (PLA).
- **Connectivity:** Banana Clips (Output/GND).

Dimensions & Placeholders

PCB Dimensions	90×110 mm
Enclosure Dimensions	$127 \times 200 \times 90$ mm

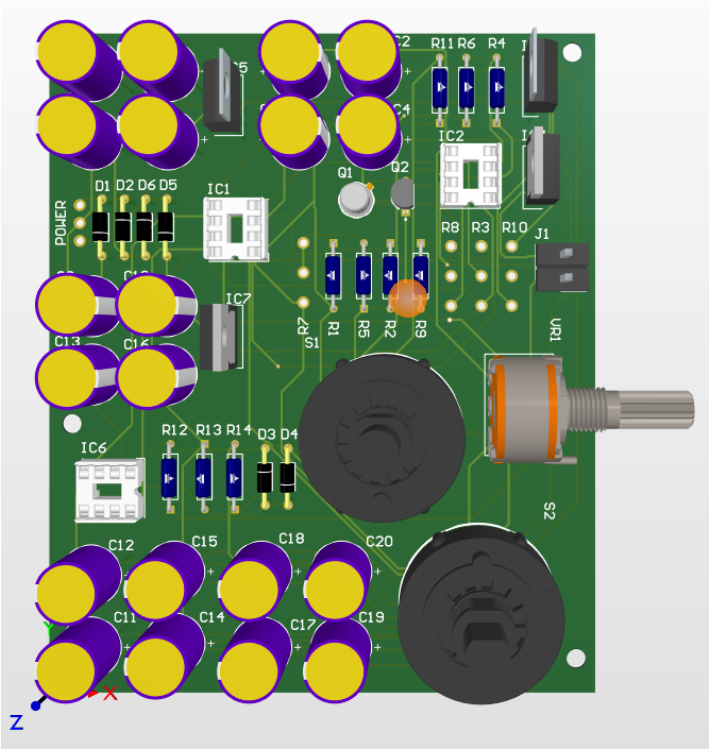

Figure 2: Internal PCB Layout

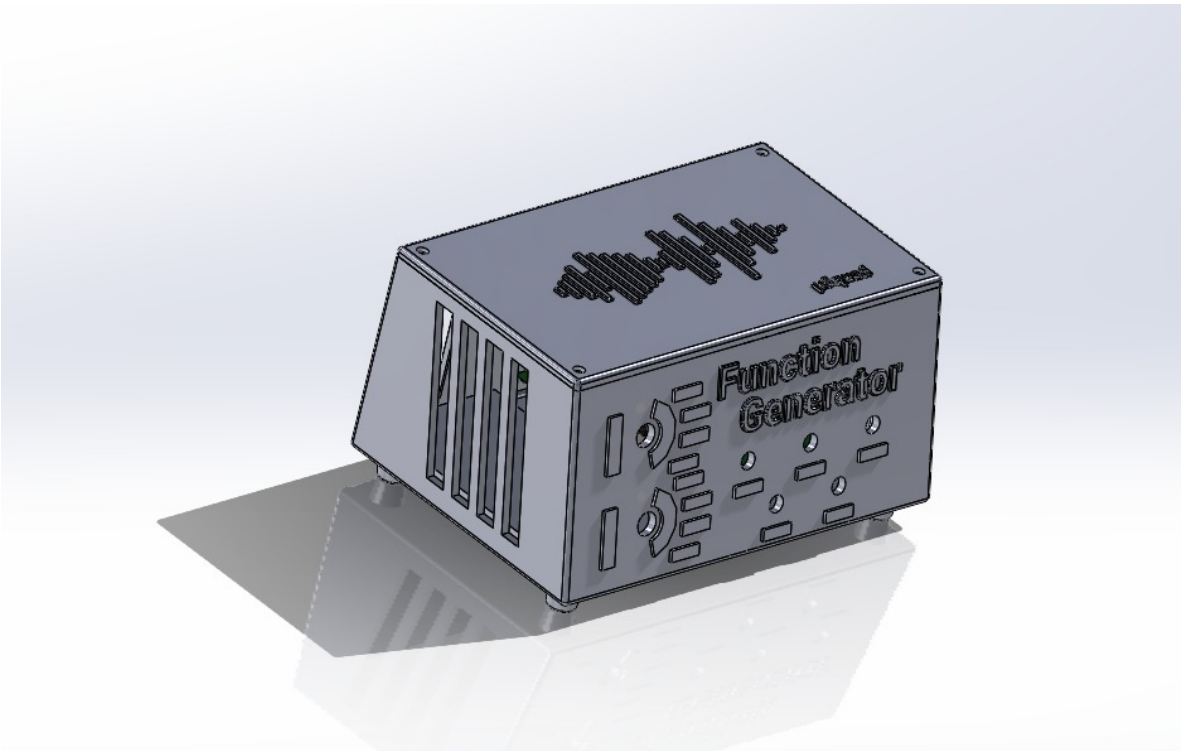

Figure 3: Enclosure Design

FRONT PANEL CONTROLS

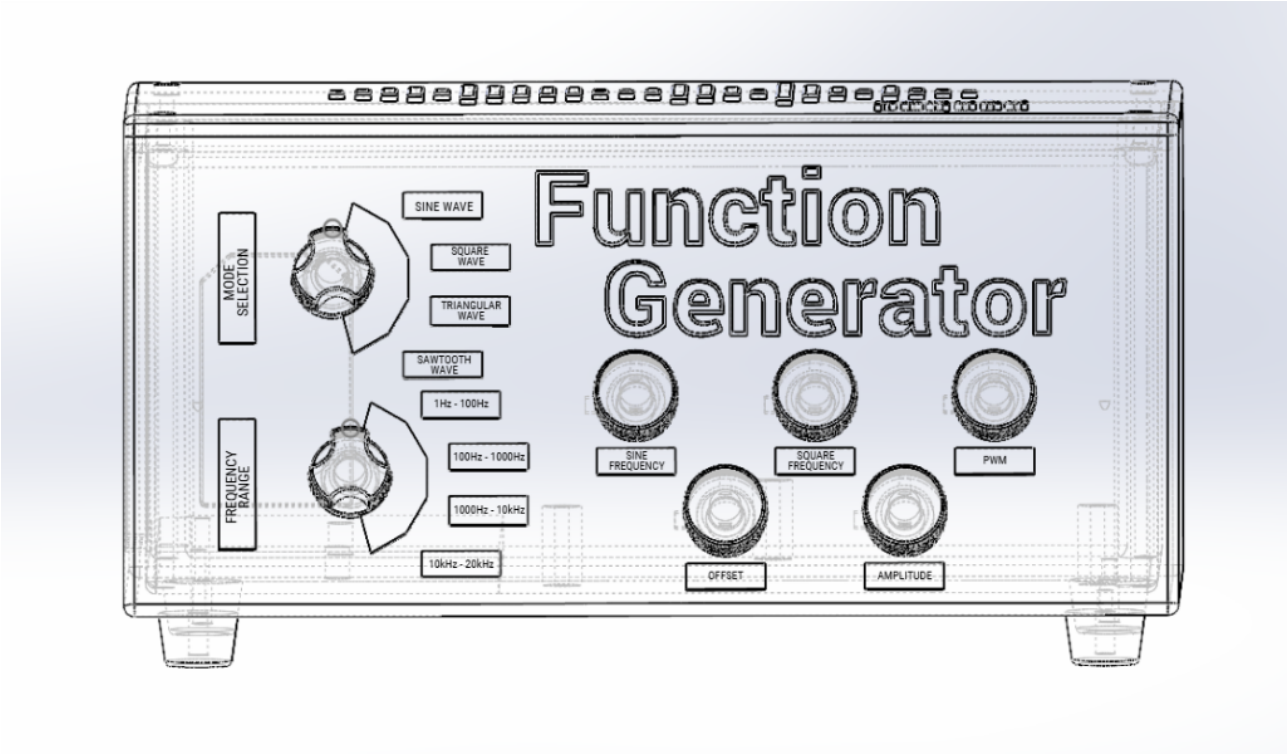

Figure 4: Front Panel Interface

Control Knob	Function Description
Mode Selection	**Rotary Switch** to select output waveform: Sine, Square, Triangular, or Sawtooth.
Frequency Range	**Rotary Switch** to select the operating frequency band: 1Hz–100Hz, 100Hz–1kHz, 1kHz–10kHz, 10kHz–20kHz.
Sine Frequency	Fine frequency adjustment potentiometer (Specific to Sine Wave).
Square Frequency	Fine frequency adjustment for Square, Triangular, and Sawtooth waves.
PWM	Pulse Width Modulation (Duty Cycle) control knob.
Offset	Adjusts the DC offset (bias) of the output signal.
Amplitude	Controls the peak-to-peak voltage level of the output.

SAFETY & HANDLING INFORMATION

- **Mains Voltage Safety:** This equipment is powered by 230V AC. To prevent the risk of electric shock, ensure the chassis remains closed while connected to the mains supply.
- **Output Protection Limits:** The signal output terminals are not protected against high external voltages. Applying potentials exceeding ± 12V to the output may cause permanent damage to the internal circuitry.